LAND & LEDGERS

HISTORICAL ECONOMY, PRICES AND MATERIALS CALIBRATION

EVIDENCE LEDGER — STEP 12 FIRST EDITION

Date: 8 October 2026 | **Area:** Dakota Territory / South Dakota and explicitly marked comparable period evidence | **Status:** Seed reference, not exhaustive price database.

Research discipline

This document makes **no unsupported universal wage, rent, commodity price or cost claim**. The source material includes dated, physical examples useful for calibrating scale but too narrow to become full market price tables. Product prices should be generated through actual supplier offers, quality/grade, credit, delivered cost, bargaining, customer willingness and local conditions; period observations guide plausible ranges.

A citation must include its date, geographic market, currency/measurement unit, grade or specification, retail/wholesale status, cash/credit terms and any freight/packing included. Treat an advertising claim, estimate or lawsuit amount differently from completed sale accounts. Convert historical units explicitly (board foot vs thousand board feet, bushel vs hundredweight, live vs dressed weight).

Evidence classes

- **Direct transaction/contract:** Dated recorded payment or contract with scope and counterparties.
- **Dated claim:** Newspaper advertisement, promotional prospectus or testimony: potentially useful but not a completed sale.
- **Period method analogue:** Physical production rate or equipment specification from contemporaneous manuals.
- **Historical synthesis/calibration:** Narrow suggested engineering/economic envelope derived from several anchors.
- **Research hold:** Price/rate missing direct historical backing; do not invent a value because a game system needs one.

Verified-source starter observations

CAL-001: Contract for finished dwelling

Observation: 4770 USD contract sum, 1882-04-03, Sioux Falls, Dakota Territory.

Context: Contract specified material+finished dwelling by Sep 1.

Evidence: Dated legal contract cited by historical work reference. **Source:** Historical Work reference PDF p.5, McCormack v Phillips [B5].

Restriction: Scope-specific contract price, not cost of every house.

CAL-002: Extras/changes to dwelling

Observation: >1000 USD reported dispute amount, 1882, Sioux Falls, Dakota Territory.

Context: Claimed extras/change work in underlying construction dispute.

Evidence: Dated legal case cited by historical work reference. **Source:** Historical Work reference PDF p.5, [B5].

Restriction: Reported over $1,000, not a universal contingency percentage.

CAL-003: Two-story frame storefront / dwelling

Observation: ~80 elapsed calendar days estimated, 1880-05 to 1880-07-20, De Smet, Dakota Territory.

Context: Dated reconstruction, May start-day uncertain.

Evidence: Regional secondary researched reconstruction. **Source:** Historical Work reference PDF p.5, [B4].

Restriction: Approximate onset assumption, not exactly 80 days established.

CAL-004: Simple three-room house

Observation: 22 calendar days as reported, 1889, North Dakota.

Context: After cellar excavation; carpenter hired.

Evidence: Contemporary first-person memoir anchor. **Source:** Historical Work reference PDF p.4, [B3].

Restriction: Not any fully finished mansion; scope is specific.

CAL-005: 40x60 plank barn shell and finishing

Observation: ~2 working weeks, period 19th c., South Dakota comparable historical barn.

Context: Journeyman carpenter plus 4 men; site/material ready.

Evidence: Regional historic preservation synthesis. **Source:** Historical Work reference PDF p.6, [B6].

Restriction: Not a universal barn timer; one type and crew.

CAL-006: Wood shingles on straight roof

Observation: ~2000 shingles per capable worker per 10-hour day, 1880, North American trade source.

Context: Fast worker; 2500 reported on straight roof.

Evidence: Period trade journal comparison. **Source:** Historical Work reference PDF p.7, [C1].

Restriction: Non-Dakota comparable method, not ordinary output for complex roof.

CAL-007: Manual loading earth in carts

Observation: 15-25 cubic yards per man per 10h day, 1900, United States engineering manual.

Context: Carts, material, hauling cycle available.

Evidence: Period engineering manual. **Source:** Historical Work reference PDF p.10, [R6].

Restriction: Do not apply beyond indicated method.

CAL-008: Rail ties in one measured section

Observation: 2640 ties per route mile, 19th c. UPRR recorded, Union Pacific historical example.

Context: Source report on defined UPRR work section.

Evidence: Government engineering report. **Source:** Historical Work reference PDF p.8, [R2].

Restriction: Comparable precedent, not universal 1870–1905 Dakota tie count.

CAL-009: Track on prepared grade

Observation: ~1 route mile per day, 1900 report, Texas, comparable U.S. rail extension.

Context: Rail/ties staged; does not include survey, bridges, grading.

Evidence: Dated period newspaper. **Source:** Historical Work reference PDF p.9, [R1].

Restriction: Not start-to-service railway duration.

CAL-010: Example nominal steer sale value

Observation: 30 USD per steer (promotional claim), 1883 promotional prospectus, Montana (not Dakota).

Context: Promoter describes a steer worth $30.

Evidence: Contemporary promotional statement via NPS. **Source:** https://www.nps.gov/grko/learn/historyculture/winter.htm.

Restriction: Promotional assertion, not verified market-clearing price or inflation-adjusted.

CAL-011: Example annual cattle care cost

Observation: 0.75 USD per animal-year (promotional claim), 1883 promotional prospectus, Montana (not Dakota).

Context: Promoter suggests cheap open-range care.

Evidence: Contemporary promotional statement via NPS. **Source:** https://www.nps.gov/grko/learn/historyculture/winter.htm.

Restriction: Optimistic sales prospectus, not true full operating cost.

CAL-012: Persons in agricultural occupations

Observation: 82857 persons employed gainfully, 1900, South Dakota.

Context: 1900 state Census occupational category counts.

Evidence: Census statistics reproduced in occupation report. **Source:** Occupation reference PDF p.10.

Restriction: Demography context, not a wage or business count; 1870 world differs.

Outstanding research datasets

- **HOLD-001 — Wage series:** Daily cash wage by trade/season, 1870-1905, Dakota county and occupation; target unit USD/day; Need local payroll, ads and census wage tables.
- **HOLD-002 — Boarding and room rents:** Rent versus full board and meals, 1870-1905, Dakota town/county; target unit USD/week or USD/month; Need newspaper ads/account ledgers with amenity scope.
- **HOLD-003 — Rail and wagon tariffs:** Freight rates by distance, cargo, route and capacity, 1870-1905, Actual origin/destination corridor; target unit USD/cwt or USD/ton-mile; Need tariff sheets waybills and carrier notices.
- **HOLD-004 — Wholesale merchandise:** Retail and regional jobber unit costs, 1870-1905, Omaha/Sioux City/Chicago vs Dakota shops; target unit USD per unit/lot; Need source grade and packaging plus freight.
- **HOLD-005 — Livestock market grades:** Cattle/hog/sheep price by weight and quality, 1870-1905, Dakota farm/terminal stockyards; target unit USD/cwt live weight; Need dated stockyard prices and weighed grades.
- **HOLD-006 — Agricultural crop yields:** Grain yield by acreage, 1870-1905, Dakota county/land/weather; target unit bushels/acre; Need survey/USDA observations and weather context.
- **HOLD-007 — Farm implement cost:** Reapers, threshers, plows, tractors, 1870-1905, Dakota dealer vs regional hub; target unit USD per machine; Need dated catalogs, supplier finance, parts and freight.
- **HOLD-008 — Sawmill/forge input yields:** Log-to-lumber, iron-to-nails throughput, 1870-1905, Local production equipment; target unit board feet/hour, units/working day; Need direct period equipment/manual analogues.
- **HOLD-009 — Deposit/lending rates:** Promissory-note rates, discounts, deposit terms, 1870-1905, Bank type, region, borrower; target unit nominal annual %; Need real contract/legal ceiling separate from quoted rates.
- **HOLD-010 — Historical food retail:** Eggs, bread, flour, butter, preserved goods, 1870-1905, Local store/settlement; target unit USD/dozen, USD/lb, USD/barrel; Need actual point-of-sale/advertising records and quality/package.
- **HOLD-011 — Construction materials:** Lumber, bricks, roofing, windows, 1870-1905, Dakota source/destination; target unit USD/board foot, per 1000 bricks, etc; Need freight-included vs direct cost.
- **HOLD-012 — Currency and coin/credit:** Cash discounts, trade credit, collection delays, 1870-1905, Bank/merchant/network; target unit USD and days; Need household and merchant ledger samples.

Implementation boundaries

1. Preserve exact authoring currency, unit, and contract terms; a dollar in a claim document is not a separately stored modern purchasing-power score.
2. Landed cost should distinguish invoice cost, actual freight, handling, deposits, tax/duty where applicable, spoilage and financed carrying costs.
3. Prices can change because supply, demand, credit, weather, quality and routes change; no historical month-based price buff is justified without underlying physical changes.
4. Worker time and crew composition determine payroll and output. A benchmark rate is never added directly to money.
5. A scenario-specific example or autonomous playtest-generated cash total does not become authoritative economic calibration.
6. Historical retail, supplier and livestock records should be stored in queryable per-item/per-unit datasets after period source verification.

Publication status: Research ledger has anchor examples, but full REF-05 economic calibration remains an open research workstream; the 12 gaps above must be populated before the specialist document is marked final.